

Doubly Fair Dynamic Pricing

A Complete, Rigorous Walk-through of
Xu, Qiao & Wang — AISTATS 2023

Aman Singh (BS2308)

Indian Statistical Institute, Kolkata

December 2023

Presentation Outline

- 1 Background & Motivation
- 2 Motivating Example
- 3 Problem Setup
- 4 The FPA Algorithm
- 5 Regret and Unfairness Analysis
- 6 Discussion and Extensions
- 7 Conclusion

- 1 Background & Motivation
- 2 Motivating Example
- 3 Problem Setup
- 4 The FPA Algorithm
- 5 Regret and Unfairness Analysis
- 6 Discussion and Extensions
- 7 Conclusion

The Classical Online Dynamic Pricing Problem

Online Pricing Protocol (per round t)

1. Customer arrives with *private* valuation y_t .
2. Seller proposes price v_t **without knowing** y_t .
3. Customer buys iff $v_t \leq y_t$:

$$\mathbf{1}_t = \mathbf{1}(v_t \leq y_t).$$

4. Seller collects revenue $r_t = v_t \cdot \mathbf{1}_t$.

Censored feedback: The seller observes only buy/no-buy, never y_t itself. The demand curve must be *learned* from binary signals.

Why is this hard?

- ▶ **Price too low** $\Rightarrow$ customer buys, but revenue is suboptimal.
- ▶ **Price too high** $\Rightarrow$ customer rejects; no revenue, no direct feedback on true y_t .
- ▶ Only the *comparison* $v_t \leq y_t$ is revealed.

This creates an **exploration–exploitation trade-off**: try varied prices to learn demand vs. exploit the best known price.

Goal: $\max \sum_{t=1}^T r_t$ by learning demand from binary feedback over T rounds.

The Fairness Problem

In modern markets, customers come from **observable groups**:

- ▶ Gender, race, age, location, . . .
- ▶ Groups may have different demand curves.
- ▶ Revenue-maximising prices may **differ** by group.

Without constraints, the seller might charge Group A systematically more than Group B — even for the *same product*. This is **price discrimination**.

Two Real-World Scenarios

Scenario 1 (Procedural):

Women browsing an online store are *offered* consistently higher average prices than men for identical items.

Scenario 2 (Substantive):

Among those who *actually buy*, women *paid* a higher average price than men.

Both are unfair — but they are **different kinds** of unfairness!

This Paper's Question

Can a seller learn to price dynamically over time so that revenue is near-optimal *and* the pricing is **doubly fair** — fair both in what is offered and in what is ultimately paid?

Two Types of Fairness: Procedural vs. Substantive

Procedural Fairness

Inspired by *procedural unconscionability* (Elfin, 1988).

Definition (informal):

The *prices offered* to both groups should be equal in expectation.

What it controls: the decision rule of the seller.

Think of it as: “equal treatment at the offer stage.”

Easy or hard? **Tractable.** Depends only on the policy, not on unknown demand.

Substantive Fairness

Inspired by *substantive unconscionability* (Elfin, 1988) — **new to this paper.**

Definition (informal):

The *prices actually accepted/paid* by buyers in both groups should be equal in expectation.

What it controls: the realized market outcome.

Think of it as: “equal outcomes among buyers.”

Easy or hard? **Hard.** Depends on *unknown* demand distributions.

Key Insight

A policy that is **simultaneously** procedurally *and* substantively fair is called **doubly fair**. This is the paper's central object of study.

Why Satisfying Both Constraints is Non-Trivial

The conflict:

- ▶ Group 1 and Group 2 may have *different* optimal prices.
- ▶ Forcing both groups to receive the same deterministic price may cost significant revenue.
- ▶ Yet with *only* deterministic prices, satisfying **both** fairness constraints simultaneously may leave no option but the same price for both groups.

The paper's resolution: Allow the seller to use **randomized pricing policies** — a distribution over prices for each group — and measure fairness *in expectation*. This unlocks strictly better doubly-fair solutions.

Why Randomization?

A single deterministic price must lie exactly at the intersection of two fairness constraints. This intersection can be:

- ▶ **Empty** (no fair price exists), or
- ▶ **A singleton** (only one price is fair, which may be revenue-poor).

A **distribution over prices** can satisfy fairness *on average* while visiting different prices for different group members — extracting more revenue.

Implementation: Random discounts, randomized coupons, promotional lotteries.

- 1 Background & Motivation
- 2 Motivating Example**
- 3 Problem Setup
- 4 The FPA Algorithm
- 5 Regret and Unfairness Analysis
- 6 Discussion and Extensions
- 7 Conclusion

Example 1: Two Groups, Three Prices

Setup

- ▶ **Two groups:** G_1 (30% of customers), G_2 (70%).
- ▶ **Three candidate prices:** $\{0.625, 0.70, 1.00\}$.
- ▶ **Acceptance rates** (probability of buying):

	Price	\$0.625	\$0.70	\$1.00
G_1 (30%)		3/5	1/2	1/2
G_2 (70%)		4/5	4/5	1/2

Expected revenue at price v for group G_e :

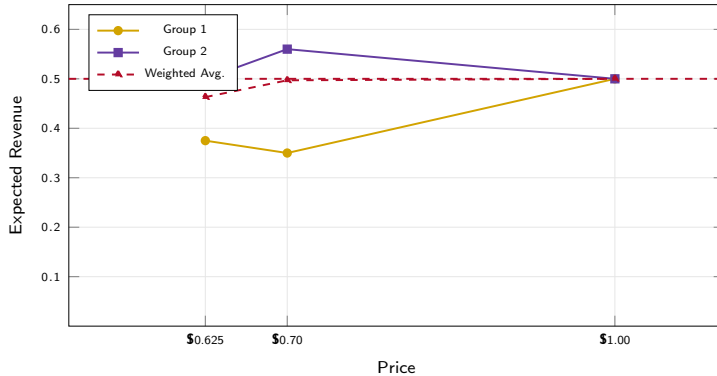
$$\text{Rev}_e(v) = v \times \text{Pr}(\text{accept in } G_e)$$

	\$0.625	\$0.70	\$1.00
G_1	0.375	0.350	0.500
G_2	0.500	0.560	0.500
Weighted avg.	0.463	0.497	0.500

Under **deterministic** fair pricing, the only way to satisfy both fairness constraints is to charge the **same price** to both groups.

Best such price: \$1.00, revenue = **\$0.50**.

Example 1: Revenue Curves



Example 1: The Optimal Randomized Policy

The Doubly-Fair Random Policy

G_1 : \$0.625 w.p. $\frac{20}{29}$, \$1.00 w.p. $\frac{9}{29}$

G_2 : \$0.70 w.p. $\frac{25}{29}$, \$1.00 w.p. $\frac{4}{29}$

Step-by-step verification:

Expected proposed price (G1):

$$\frac{20}{29} \times 0.625 + \frac{9}{29} \times 1.00 = \frac{12.5+9}{29} = \frac{21.5}{29} = \frac{43}{58}$$

Expected proposed price (G2):

$$\frac{25}{29} \times 0.70 + \frac{4}{29} \times 1.00 = \frac{17.5+4}{29} = \frac{21.5}{29} = \frac{43}{58}$$

✓ **Procedural Fairness: Both equal $\frac{43}{58}$!**

Expected accepted price (G1):

$$\begin{aligned} \text{Numerator (revenue): } & \frac{20}{29}(0.625)(0.6) + \frac{9}{29}(1)(0.5) \\ &= \frac{20 \times 0.375 + 9 \times 0.5}{29} = \frac{7.5+4.5}{29} = \frac{12}{29} \end{aligned}$$

Denominator (accept rate):

$$\frac{20}{29}(0.6) + \frac{9}{29}(0.5) = \frac{12+4.5}{29} = \frac{16.5}{29}$$

$$\Rightarrow \frac{12/29}{16.5/29} = \frac{12}{16.5} = \frac{8}{11}$$

Expected accepted price (G2):

$$\text{Numerator: } \frac{25}{29}(0.7)(0.8) + \frac{4}{29}(1)(0.5) = \frac{14+2}{29} = \frac{16}{29}$$

$$\text{Denominator: } \frac{25}{29}(0.8) + \frac{4}{29}(0.5) = \frac{20+2}{29} = \frac{22}{29}$$

$$\Rightarrow \frac{16}{22} = \frac{8}{11}$$

✓ **Substantive Fairness: Both equal $\frac{8}{11}$!**

Expected revenue:

$$0.3 \times \frac{12}{29} + 0.7 \times \frac{16}{29} = \frac{3.6+11.2}{29} = \frac{14.8}{29} \approx \frac{74}{145}$$

$\frac{74}{145} \approx 0.510 > 0.50 = \text{best fixed-price!}$

Key Takeaways from Example 1

What the Example Proves

1. **Deterministic doubly-fair pricing** forces the same price for both groups; best fixed-price revenue = \$0.50.
2. **A randomized policy** can satisfy both fairness constraints *in expectation* while achieving revenue $\frac{74}{145} > 0.50$.
3. The displayed policy is in fact the **globally optimal doubly-fair randomized policy** for this instance (proved in Appendix B.3 of the paper).
4. Randomization is **mathematically necessary**, not just a convenience — the optimal fair solution may lie *outside* the set of deterministic prices.

Mental Model

Think of the policy as a **lottery** the seller runs for each group. Each group gets the same average price and average accepted price, but the lottery tickets differ — allowing revenue extraction without systematic discrimination.

The Paper's Challenge

In the *online* setting, the seller does **not know** the acceptance probabilities $F_1(\cdot), F_2(\cdot)$. They must be *learned* from binary feedback. The algorithm must simultaneously **explore** to estimate demand and **exploit** to maximize revenue while **maintaining fairness**.

- 1 Background & Motivation
- 2 Motivating Example
- 3 Problem Setup**
- 4 The FPA Algorithm
- 5 Regret and Unfairness Analysis
- 6 Discussion and Extensions
- 7 Conclusion

The Environment

- ▶ **Time horizon:** T rounds.
- ▶ **Two groups:** G_1, G_2 ; customer is from G_e , $e \in \{1, 2\}$.
- ▶ **Group share:** $q = \Pr(\text{customer is from } G_1)$.
- ▶ **Price set:** finite, ordered, known:

$$V = \{v_1 < v_2 < \dots < v_d\} \subset (0, 1].$$

- ▶ **Valuation** y_t^e : drawn i.i.d. from unknown D_e .
- ▶ **Feedback:** $\mathbf{1}_t^e = \mathbf{1}(v_t^e \leq y_t^e)$ (buy/no-buy).
- ▶ **Revenue:** $r_t^e = v_t^e \cdot \mathbf{1}_t^e$.

Policies and the Simplex

A **pricing policy** is a pair of distributions over V :

$$\pi = (\pi^1, \pi^2), \quad \pi^e \in \Delta^d,$$

where $\Delta^d = \{x \in \mathbb{R}_{\geq 0}^d : \|x\|_1 = 1\}$ is the probability simplex.

$\pi^e(i)$ = probability of offering price v_i to group G_e .

Special cases:

- ▶ Fixed price v_i : $\pi^e = e_i$ (standard basis vector).
- ▶ Mixed policy: any interior point of Δ^d .

Useful notation: Let $\mathbf{v} = [v_1, \dots, v_d]^\top$, $\mathbf{1} = [1, \dots, 1]^\top \in \mathbb{R}^d$, and $[d] = \{1, \dots, d\}$.

Definition: Acceptance Probability

For group G_e and price v_i , define:

$$F_e(i) = \Pr_{D_e}[y_t^e \geq v_i].$$

This is the probability that a customer from G_e buys at price v_i .

Monotonicity: Since prices are ordered $v_1 < \dots < v_d$:

$$F_e(1) \geq F_e(2) \geq \dots \geq F_e(d) > 0.$$

Matrix form:

$$F_e = \text{diag}(F_e(1), F_e(2), \dots, F_e(d)) \in \mathbb{R}^{d \times d}.$$

F_e is a diagonal matrix of acceptance probabilities.

Example (from Example 1)

$V = \{0.625, 0.70, 1.00\}$, so $d = 3$.

For Group 1:

$$F_1 = \text{diag}(3/5, 1/2, 1/2)$$

For Group 2:

$$F_2 = \text{diag}(4/5, 4/5, 1/2)$$

Critical: F_1 and F_2 are *unknown to the seller*.
The entire algorithm is designed to learn these from binary buy/no-buy feedback!

Four Key Quantities for Group G_e

Given policy π^e and acceptance matrix F_e , the four fundamental quantities are:

1. Expected Proposed Price

$$\mathbb{E}[v_t^e] = \sum_{i=1}^d v_i \pi^e(i) = \mathbf{v}^\top \pi^e$$

What the seller offers on average to G_e .

2. Expected Revenue

$$\mathbb{E}[r_t^e] = \sum_{i=1}^d v_i F_e(i) \pi^e(i) = \mathbf{v}^\top F_e \pi^e$$

Price $\times$ acceptance prob., average over policy.

3. Expected Acceptance Rate

$$\Pr(\text{accept} \mid G_e) = \sum_{i=1}^d F_e(i) \pi^e(i) = \mathbf{1}^\top F_e \pi^e$$

Fraction of customers from G_e who buy.

4. Expected Accepted Price (crucial!)

$$\mathbb{E}[v \mid \text{accept}, G_e] = \frac{\mathbf{v}^\top F_e \pi^e}{\mathbf{1}^\top F_e \pi^e}$$

*Average price paid by buyers in G_e . This is a **ratio**!*

Why ratio? Among customers who accept, prices are selection-biased: cheaper prices get accepted more often. So the accepted price $\neq$ offered price. This ratio causes the non-convexity in substantive fairness.

Definition 2 — Expected Revenue

For policy $\pi = (\pi^1, \pi^2)$, the expected total revenue (weighted by population) is:

$$R(\pi; F_1, F_2) = q \cdot \mathbf{v}^\top F_1 \pi^1 + (1 - q) \cdot \mathbf{v}^\top F_2 \pi^2$$

Definition 3 — Procedural Unfairness

Absolute difference in expected **proposed** prices:

$$U(\pi) = |\mathbf{v}^\top \pi^1 - \mathbf{v}^\top \pi^2| = |\mathbf{v}^\top (\pi^1 - \pi^2)|$$

Tractable: depends only on π , not on unknown F_e .

$\Pi := \{\pi \mid U(\pi) = 0\}$ = all procedurally fair policies.

Definition 4 — Substantive Unfairness

Absolute difference in expected **accepted** prices:

$$S(\pi; F_1, F_2) = \left| \frac{\mathbf{v}^\top F_1 \pi^1}{\mathbf{1}^\top F_1 \pi^1} - \frac{\mathbf{v}^\top F_2 \pi^2}{\mathbf{1}^\top F_2 \pi^2} \right|$$

Hard: depends on *unknown* F_1, F_2 .

Non-linear (ratio of expectations) $\Rightarrow$ non-convex feasible set!

Checking the Two Fairness Conditions

For the randomized policy from Example 1:

$$\pi^{1*} = \left(\frac{20}{29}, 0, \frac{9}{29} \right), \quad \pi^{2*} = \left(\frac{25}{29}, 0, \frac{4}{29} \right).$$

Procedural Fairness

$$\mathbf{v}^\top \pi^1 = \mathbf{v}^\top \pi^2 = \frac{43}{58}$$

Therefore,

$$U(\pi) = 0.$$

Substantive Fairness

$$\frac{\mathbf{v}^\top F_1 \pi^1}{\mathbf{1}^\top F_1 \pi^1} = \frac{\mathbf{v}^\top F_2 \pi^2}{\mathbf{1}^\top F_2 \pi^2} = \frac{8}{11}$$

Therefore,

$$S(\pi; F_1, F_2) = 0.$$

The Optimal Doubly-Fair Policy π^*

$$\pi^* = \arg \max_{\pi=(\pi^1, \pi^2) \in \Pi} R(\pi; F_1, F_2) \quad \text{s.t.} \quad U(\pi) = 0, \quad S(\pi; F_1, F_2) = 0$$

Feasibility: Trivially feasible — same fixed price v_i to both groups always satisfies $U = 0$, $S = 0$.

Hardness: Even knowing F_1, F_2 exactly, finding π^* is a **non-convex, non-quasi-convex** optimization problem (substantive fairness constraint is a ratio).

Definition 5 — Cumulative Regret

For algorithm $\mathcal{A}$ producing policies π_t :

$$\text{Reg}_T(\mathcal{A}) = \sum_{t=1}^T [R(\pi^*; F_1, F_2) - R(\pi_t; F_1, F_2)]$$

Measures revenue loss compared to the **fair** optimum. $\text{Reg}_T(\mathcal{A}) = o(T)$ indicates eventual learning.

Cumulative Substantive Unfairness

$$S_T(\mathcal{A}) = \sum_{t=1}^T S(\pi_t; F_1, F_2)$$

Total accumulated substantive unfairness over all rounds.

Goal: Both Reg_T and S_T should be $o(T)$ — vanishing *on average*.

Assumption 1 — Minimum Acceptance Probability

There exists $F_{\min} > 0$ such that for $e = 1, 2$:

$$F_e(d) \geq F_{\min}.$$

Even the highest price v_d has a positive acceptance probability.

Why needed:

- ▶ Ensures the *denominator* $\mathbf{1}^\top F_e \pi^e \geq F_{\min}$ in the accepted-price ratio is bounded away from zero.
- ▶ Gives $S(\pi, F_1, F_2)$ a Lipschitz property essential for analysis.
- ▶ Can be enforced by reducing v_d if necessary.

Assumption 2 — Price Grid Size

Treat d as independent of T . Further assume:

$$d = O(T^{1/3}).$$

Why needed: Separates the dependence on the action-space size d from the time horizon T in the regret bound. Allows the paper to show optimality with respect to T .

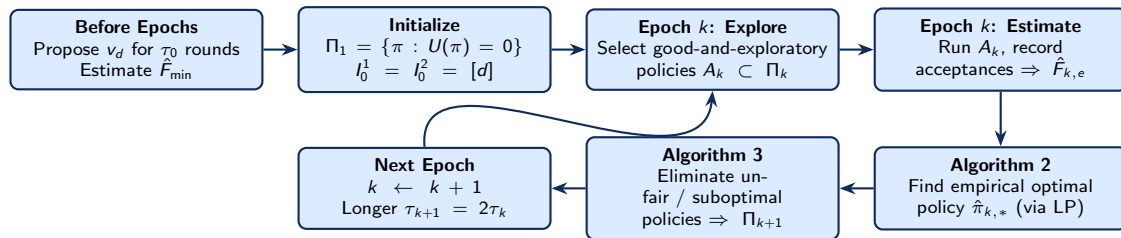
Algorithm's Target

Design $\mathcal{A}$ achieving:

- ▶ $\text{Reg}_T(\mathcal{A}) = \tilde{O}(\sqrt{T})$
- ▶ $U(\pi_t) = 0$ **for all** t (exact!)
- ▶ $S_T(\mathcal{A}) = \tilde{O}(\sqrt{T})$

- 1 Background & Motivation
- 2 Motivating Example
- 3 Problem Setup
- 4 The FPA Algorithm**
- 5 Regret and Unfairness Analysis
- 6 Discussion and Extensions
- 7 Conclusion

Algorithm Overview: Fairly Pricing Algorithm (FPA)



Core idea: Learn a rough market model $\rightarrow$ optimize on it $\rightarrow$ prune bad policies $\rightarrow$ repeat. Doubling epochs improve estimates while maintaining exact procedural fairness.

Before-Epoch Procedure

Initialize: $M_{0,e} = N_{0,e} = 0$ for $e = 1, 2$.

For $t = 1, 2, \dots, \tau_0 = 2 \log T \cdot \log \frac{16}{\epsilon}$:

1. Observe customer group $e_t \in \{1, 2\}$.
2. Set $M_{0,e_t} += 1$.
3. Propose highest price v_d .
4. If accepted: $N_{0,e_t} += 1$.

Output:

$$\hat{F}_{\min} = \min \left\{ \frac{N_{0,1}}{2M_{0,1}}, \frac{N_{0,2}}{2M_{0,2}} \right\}$$

Why v_d ? Why the factor $1/2$?

Why propose v_d ? The highest price has the lowest acceptance probability. If $F_e(d) > 0$, then all lower prices satisfy $F_e(i) \geq F_e(d)$. So $F_e(d)$ is the tightest lower bound.

Why divide by 2? Conservative safety margin. The raw ratio $N_{0,e}/(M_{0,e})$ estimates $F_e(d)$, but with high probability the true value exceeds half the estimate. This prevents over-estimating $F_{\min}$, which would later cause instability in the accepted-price ratios.

Why $\tau_0 = O(\log T)$? Enough rounds for concentration, small overhead.

Doubling Epochs: Balancing Exploration and Exploitation



Epoch Length Schedule

Epochs $k = 1, 2, \dots$, each with length:

$$\tau_k = O(\sqrt{T} \cdot 2^k)$$

Epoch k is **twice as long** as epoch $k - 1$.

Why doubling?

- ▶ Early epochs are short: more exploration uncertainty.
- ▶ Later epochs are longer: better estimates enable exploitation.
- ▶ Total epochs: $K = O(\log T)$; epoch lengths telescope.

Key trade-off: In epoch k , FPA uses a fixed policy set Π_k . Longer epochs provide more data for estimating F_e and pruning policies, but they adapt more slowly if Π_k is imperfect — so elimination must be conservative.

Selecting A_k (Good-and-Exploratory)

Initialize: $A_k = \emptyset$, $I_k^e = I_{k-1}^e$ for $e = 1, 2$.

For each group $e \in \{1, 2\}$ **and price index** $i \in I_{k-1}^e$:

1. Find the policy in Π_k that **maximizes** the probability of proposing v_i to G_e :

$$\tilde{\pi}_{k,i,e} = \arg \max_{\pi \in \Pi_k} \pi^e(i)$$

2. **If** $\tilde{\pi}_{k,i,e}^e(i) \geq \frac{1}{\sqrt{T}}$:
Add $\tilde{\pi}_{k,i,e}$ to A_k .

3. **Else:** Remove i from I_k^e .

Output: A_k and updated I_k^1, I_k^2 .

Intuition

Problem: A revenue-maximizing policy may almost never propose some prices v_i , leaving their acceptance probabilities unestimated.

Solution: For each price v_i , find a *fair* policy that proposes it as often as possible. If even this cannot achieve probability $\geq 1/\sqrt{T}$, then v_i is so rarely used by any fair policy that it will not affect the optimum — safely ignored.

Effect: $|A_k| \leq 2d$, so exploring all prices costs at most $2d$ different policies per epoch.

Acceptance Probability Estimation

Initialize: $M_{k,e}(i) = N_{k,e}(i) = 0$.

For each policy $\pi \in A_k$:

- ▶ Run π for $\tau_k/|A_k|$ rounds.
- ▶ For each proposed price v_i in G_e : $M_{k,e}(i) += 1$.
- ▶ For each accepted price v_i in G_e : $N_{k,e}(i) += 1$.

For $e = 1, 2$ **and** $i \in I_k^e$:

$$\bar{F}_{k,e}(i) = \max \left\{ \frac{N_{k,e}(i)}{M_{k,e}(i)}, \hat{F}_{\min} \right\}$$

For $i \notin I_k^e$: $\bar{F}_{k,e}(i) = \hat{F}_{\min}$.

Matrix: $\hat{F}_{k,e} = \text{diag}(\bar{F}_{k,e}(1), \dots, \bar{F}_{k,e}(d))$.

Numerical Example

Suppose for G_1 , price v_2 after an epoch:

- ▶ $M_{k,1}(2) = 40$ (proposed 40 times)
- ▶ $N_{k,1}(2) = 24$ (accepted 24 times)

Raw estimate: $24/40 = 0.60$.

If $\hat{F}_{\min} = 0.10$: store $\bar{F}_{k,1}(2) = 0.60$.

But if sample size were small ($N = 3, M = 10$):
Raw estimate = 0.30; store $\max(0.30, 0.10) = 0.30$.

Why floor at $\hat{F}_{\min}$? Prevents the denominator in the accepted-price ratio from being too small due to sampling noise, keeping the optimization numerically stable and the Lipschitz argument valid.

Why the Optimization is Hard

At epoch k , we want to find:

$$\hat{\pi}_{k,*} = \arg \max_{\pi \in \Pi_k} R(\pi, \hat{F}_{k,1}, \hat{F}_{k,2})$$

subject to:

$$S(\pi, \hat{F}_{k,1}, \hat{F}_{k,2}) \leq \delta_{k,s}$$

The constraint $S(\cdot) \leq \delta_{k,s}$ involves a **ratio**:

$$\left| \frac{\mathbf{v}^\top F_1 \pi^1}{\mathbf{1}^\top F_1 \pi^1} - \frac{\mathbf{v}^\top F_2 \pi^2}{\mathbf{1}^\top F_2 \pi^2} \right| \leq \delta_{k,s}$$

This is **non-convex** in π . Linear programming does not directly apply.

The w -Linearization Trick

Introduce a scalar $w \in [0, 1/F_{\min}]$ representing the **common accepted-price level**.

If we **fix** w , the constraints become **linear**:

$$\mathbf{v}^\top F_1 \pi^1 = w \cdot \mathbf{1}^\top F_1 \pi^1$$

$$\mathbf{v}^\top F_2 \pi^2 \geq (w - \delta_{k,s}) \cdot \mathbf{1}^\top F_2 \pi^2$$

$$\mathbf{v}^\top F_2 \pi^2 \leq (w + \delta_{k,s}) \cdot \mathbf{1}^\top F_2 \pi^2$$

For fixed w , all constraints are **linear in π** !

Algorithm: Search over a grid $w_\ell = \ell \cdot \epsilon$ where $\epsilon = \delta_{k,s}/2$, solve an LP at each w_ℓ , return the best.

Why works? The objective & constraints are Lipschitz in w , so grid approximation introduces error only $O(\delta_{k,s})$, which stays within tolerance. The number of LP solves is $O(1/\delta_{k,s})$ — polynomial in T & d .

Algorithm 2: Empirical Optimal Policy Oracle

Algorithm 2 — Empirical Optimal

Input: $\hat{F}_{k,1}$, $\hat{F}_{k,2}$, Π_k , $\hat{F}_{\min}$, $\delta_{k,s}$, step $\epsilon = \delta_{k,s}/2$.

Initialize: $\hat{\pi}_{k,*} \leftarrow$ any $\pi \in \Pi_k$.

For $\ell = 0, 1, 2, \dots$: $w_\ell = \ell \cdot \epsilon$. **Break if** $w_\ell > 1/\hat{F}_{\min}$.

Solve LP:

$$\hat{\pi}_{k,\ell,*} = \arg \max_{\pi \in \Pi_k} R(\pi, \hat{F}_{k,1}, \hat{F}_{k,2})$$

$$\text{s.t. } \begin{cases} \mathbf{v}^\top F_1 \pi^1 = w_\ell \cdot \mathbf{1}^\top F_1 \pi^1 \\ \mathbf{v}^\top F_2 \pi^2 \geq (w_\ell - \delta_{k,s}) \cdot \mathbf{1}^\top F_2 \pi^2 \\ \mathbf{v}^\top F_2 \pi^2 \leq (w_\ell + \delta_{k,s}) \cdot \mathbf{1}^\top F_2 \pi^2 \end{cases}$$

If $R(\hat{\pi}_{k,\ell,*}, \dots) > R(\hat{\pi}_{k,*}, \dots)$: update $\hat{\pi}_{k,*} \leftarrow \hat{\pi}_{k,\ell,*}$.

Return $\hat{\pi}_{k,*}$.

Intuition: The w -Dial

Imagine a dial labeled “fair accepted price level.”

- ▶ Turn the dial to $w = 0.6$: find the best revenue policy that makes both groups' buyers pay average ≈ 0.6 .
- ▶ Turn to $w = 0.7$: repeat.
- ▶ ... continue up to $w = 1/F_{\min}$.
- ▶ Return the overall best solution found.

Each LP is linear and solvable in polynomial time.

Total LPs solved: $O(1/\delta_{k,s})$ per epoch;
 $O(\log T)$ epochs.

$\Rightarrow$ **Polynomial time overall.**

Algorithm 3: Policy Elimination Oracle

Algorithm 3 — Policy Elimination

Input: $\hat{F}_{k,e}, \Pi_k, \hat{F}_{\min}, \delta_{k,s}, \delta_{k,r}, L, \hat{\pi}_{k,*}, \epsilon = \delta_{k,s}/2$.

Initialize: $\Pi_{k+1} = \emptyset$.

For $\ell = 0, 1, 2, \dots$: $w_\ell = \ell \cdot \epsilon$. **Break if** $w_\ell > 1/\hat{F}_{\min}$.

Find feasible set Π_{k+1,w_ℓ} : all $\pi \in \Pi_k$ satisfying:

$$\begin{cases} \mathbf{v}^\top F_1 \pi^1 = w_\ell \cdot \mathbf{1}^\top F_1 \pi^1 \\ \mathbf{v}^\top F_2 \pi^2 \geq (w_\ell - \delta_{k,s}) \cdot \mathbf{1}^\top F_2 \pi^2 \\ \mathbf{v}^\top F_2 \pi^2 \leq (w_\ell + \delta_{k,s}) \cdot \mathbf{1}^\top F_2 \pi^2 \\ R(\pi, \hat{F}_{k,1}, \hat{F}_{k,2}) \geq R(\hat{\pi}_{k,*}, \hat{F}_{k,1}, \hat{F}_{k,2}) \\ \quad - \delta_{k,r} - L \cdot \delta_{k,s} \end{cases}$$

Set $\Pi_{k+1} \leftarrow \Pi_{k+1} \cup \Pi_{k+1,w_\ell}$.

Return Π_{k+1} .

Why Two Elimination Criteria?

A policy is **eliminated** if it satisfies either:

1. **Too unfair:**

$$S(\pi, \hat{F}_{k,1}, \hat{F}_{k,2}) > \delta_{k,s}$$

(Accepted-price gap exceeds threshold.)

2. **Too sub-optimal:**

$$R(\pi) < R(\hat{\pi}_{k,*}) - \delta_{k,r} - L \cdot \delta_{k,s}$$

where:

- $\delta_{k,r}$: revenue estimation error buffer.
- $L \cdot \delta_{k,s}$: max revenue gain from relaxing substantive fairness by $\delta_{k,s}$.

Conservative: True optimum π^* is never eliminated!

Algorithm 1: Fairly Pricing Algorithm (FPA)

Input: horizon T , price set V , error prob. ϵ , constant L , proportion q .

1. **Before Epochs:** Propose v_d for $\tau_0 = 2 \log T \log(16/\epsilon)$ rounds; output $\hat{F}_{\min}$.
2. **Initialization:** $\Pi_1 = \Pi = \{\pi : U(\pi) = 0\}$; $I_0^1 = I_0^2 = [d]$.
3. **For epoch** $k = 1, 2, \dots$:

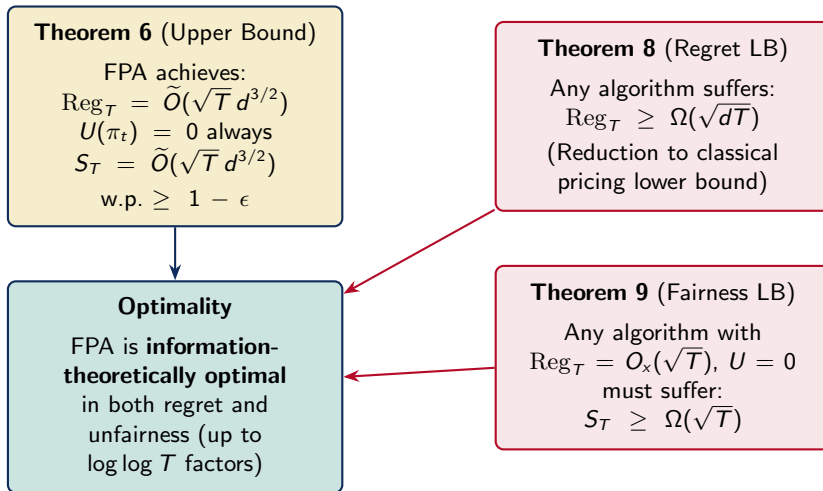
- (a) Set epoch length τ_k , tolerances $\delta_{k,r}, \delta_{k,s}$.
- (b) Select good-and-exploratory policies $A_k \subset \Pi_k$.
- (c) Estimate $\hat{F}_{k,e}$ by running each policy in A_k .
- (d) Find empirical optimal:

$$\hat{\pi}_{k,*} = \arg \max_{\pi \in \Pi_k} R(\pi, \hat{F}_{k,1}, \hat{F}_{k,2}) \quad \text{s.t.} \quad S(\pi, \hat{F}_{k,1}, \hat{F}_{k,2}) \leq \delta_{k,s}.$$

- (e) Update candidate set: eliminate policies with $S(\pi) > \delta_{k,s}$ or $R(\pi) < R(\hat{\pi}_{k,*}) - \delta_{k,r} - L \cdot \delta_{k,s}$.

Complexity: FPA is **oracle-efficient**: $O(\log T)$ epochs, $O(1/\delta_{k,s})$ LPs per epoch, polynomial total time in T , d , and $1/\epsilon$.

- 1 Background & Motivation
- 2 Motivating Example
- 3 Problem Setup
- 4 The FPA Algorithm
- 5 Regret and Unfairness Analysis**
- 6 Discussion and Extensions
- 7 Conclusion



Theorem 6 (Xu, Qiao, Wang 2023)

FPA guarantees, with probability at least $1 - \epsilon$:

$$\text{Reg}_T(\text{FPA}) = O\left(\sqrt{T} d^{3/2} \frac{\log d \log T}{\epsilon}\right)$$

$$U(\pi_t) = 0 \quad \text{for all } t = 1, 2, \dots, T \quad (\text{exactly!})$$

$$S_T(\text{FPA}) = O\left(\sqrt{T} d^{3/2} \frac{\log d \log T}{\epsilon}\right)$$

Bottom line: FPA simultaneously learns revenue-optimal prices and controls both fairness notions at the optimal $\sqrt{T}$ scale.

Theorem 6: What the Guarantee Means

What this says

- ▶ Average regret per round = $O(1/\sqrt{T}) \rightarrow 0$. The algorithm becomes asymptotically revenue-optimal.
- ▶ Procedural fairness is *exact*, not just approximate.
- ▶ Substantive unfairness also vanishes at the same $\sqrt{T}$ rate.
- ▶ Fairness constraints add **no informational obstacle** to learning.

Remark 7

The $\tilde{O}(\sqrt{T} \log \log T)$ order matches the estimation error $O(1/\sqrt{T})$ per round. These upper bounds are **tight up to $O(\log \log T)$ factors**, as shown by Theorems 8 and 9 below.

Implication: The fairness constraints do **not** force the algorithm to learn more slowly — the fundamental statistical difficulty is the same as unconstrained pricing.

Proof by Induction on Epoch k

Induction hypothesis: $\pi^* \in \Pi_k$ (the true optimal fair policy is in the candidate set).

Base case $k = 1$: $\Pi_1 = \Pi$, trivially contains π^* .

Inductive step ($k \rightarrow k + 1$):

- 1. Concentration bounds:** With high probability, $|\hat{F}_{k,e}(i) - F_e(i)| \leq \tilde{O}(1/\sqrt{\tau_k})$ for all i, e .
- 2. Error propagation:** Derive bounds on estimation error for $R(\pi, \hat{F})$ and $S(\pi, \hat{F})$ for all $\pi \in \Pi_k$.
- 3. Key:** $\pi^* \in \Pi_{k+1}$ (elimination does not remove π^*).
- 4. Regret per epoch:** Any π surviving in Π_{k+1} has $R(\pi) \geq R(\pi^*) - \tilde{O}(\delta_{k,r})$.
- 5. Sum over epochs:** Total regret = $\sum_k \tau_k \cdot \tilde{O}(\delta_{k,r}) = \tilde{O}(\sqrt{T})$.

Why π^* Is Never Eliminated

The elimination threshold includes two buffers:

- ▶ $\delta_{k,r}$: accounts for estimation error in R .
- ▶ $L \cdot \delta_{k,s}$: accounts for the possible revenue increase when allowing $S(\pi) \leq \delta_{k,s}$ instead of $S(\pi) = 0$.

Set conservatively so that true $R(\pi^*)$ and $S(\pi^*) = 0$ are within the threshold. Hence π^* always passes the test. **This is the core correctness argument.**

Analogy: Like a basketball tournament that keeps a team in until it has been *convincingly* beaten — never eliminated on a bad day.

Theorem 8: Regret Lower Bound

Theorem 8 (Regret Lower Bound)

Assume $d \leq T^{1/3}$. For any algorithm for the online two-group fair pricing problem:

$$\text{Reg}_T(\mathcal{A}) \geq \Omega(\sqrt{dT})$$

Proof Idea: Reduction

Consider the special case where the two groups are **identical**:

$$F_1(i) = F_2(i) \text{ for all } i \in [d], \quad q = 0.5.$$

Any policy with $\pi^1 = \pi^2$ is both procedurally and substantively fair (trivially). The fair pricing problem reduces to **ordinary online pricing with d prices**.

Apply the bandit-style lower bound of [Auer et al. \(2002\)](#): any algorithm for d -action online pricing with binary feedback suffers $\Omega(\sqrt{dT})$ regret.

Since the fair problem contains the ordinary problem as a special case, it cannot be easier. $\square$

Theorem 8: Interpreting the $\sqrt{dT}$ Lower Bound

Why $\sqrt{dT}$ and Not Just $\sqrt{T}$?

More prices = harder to learn.

With d candidate prices, the algorithm must distinguish which is best. Each additional price is like a new arm in a multi-armed bandit:

- ▶ $d = 2$: relatively easy.
- ▶ $d = 100$: need to try many prices, accumulate more regret during exploration.

What the Bound Means

The $\sqrt{dT}$ lower bound captures the unavoidable cost of exploration over a larger price menu.

- ▶ It applies even in the easiest fair case: two identical groups.
- ▶ Therefore fairness constraints do not remove the ordinary pricing hardness.
- ▶ FPA's upper bound $\tilde{O}(\sqrt{T} d^{3/2})$ is tight in T , though not in its dependence on d .

Theorem 9: Substantive Unfairness is Unavoidable

Theorem 9 (Substantive Unfairness Lower Bound)

For any constant $C_x > 0$, there exists $C_u > 0$ such that any algorithm with cumulative regret $\leq C_x \sqrt{T}$ and zero procedural unfairness must suffer:

$$S_T(\mathcal{A}) \geq C_u \sqrt{T}$$

This Theorem is Fundamentally Different from Theorem 8

Theorem 8 says: *learning is hard*. Any algorithm pays $\Omega(\sqrt{T})$ regret.

Theorem 9 says: *even a nearly optimal algorithm must be somewhat substantively unfair*. The unfairness cost is an **information-theoretic necessity** for any algorithm with optimal regret — not a flaw in FPA.

Together: FPA is simultaneously near-optimal in **both** regret and substantive unfairness.

Construction of Two Hard Instances

Setting A: Exactly Example 1 from the introduction.

Setting B: Same as Setting A, except acceptance rates of \$0.70 (for G_1) and \$1.00 (for both groups) are now $(0.5 - \zeta)$ for small $\zeta > 0$.

Key properties:

- ▶ The optimal fair policy π^* **differs** between Settings A and B.
- ▶ The settings are **statistically nearly identical** — same binary feedback distribution, shifted by $O(\zeta)$.
- ▶ Distinguishing A from B requires $\Omega(1/\zeta^2)$ observations.

The Indistinguishability Argument

Any algorithm faces a dilemma:

Option 1: Commit early to Setting A's policy.

- ▶ If truly in B: uses wrong policy $\Rightarrow \Omega(\sqrt{T})$ substantive unfairness.

Option 2: Try to distinguish A from B first.

- ▶ Needs $\Omega(1/\zeta^2)$ observations.
- ▶ During that time: uses sub-optimal policy $\Rightarrow \Omega(1/\zeta^2 \cdot \text{gap})$ regret.
- ▶ With tight regret budget: $\Omega(\sqrt{T})$ unfairness unavoidable.

Option 3: Avoid committing entirely.

- ▶ Suffers even larger regret or unfairness.

In all cases: $S_T \geq \Omega(\sqrt{T})$. $\square$

Information-Theoretic Optimality: Putting It Together

Metric	FPA (Upper Bound)	Lower Bound (any algorithm)
Cumulative Regret	$\tilde{O}(\sqrt{T} d^{3/2})$	$\Omega(\sqrt{dT})$
Procedural Unfairness	$\mathbf{0}$ (exact)	—
Cumulative Subst. Unfairness	$\tilde{O}(\sqrt{T} d^{3/2})$	$\Omega(\sqrt{T})$

Conclusion on Optimality

- ▶ Regret: FPA matches $\Omega(\sqrt{dT})$ lower bound **up to** $\log \log T$ factors w.r.t. T . (*The $d^{3/2}$ vs $d^{1/2}$ gap in d is a current limitation.*)
- ▶ Substantive unfairness: FPA matches $\Omega(\sqrt{T})$ lower bound up to $\log \log T$ factors.
- ▶ **Procedural fairness is exact at all times** — a remarkable guarantee given that substantive fairness must be learned.
- ▶ FPA is the **first** dynamic pricing algorithm achieving both fairness constraints simultaneously with provable near-optimal rates.

- 1 Background & Motivation
- 2 Motivating Example
- 3 Problem Setup
- 4 The FPA Algorithm
- 5 Regret and Unfairness Analysis
- 6 Discussion and Extensions**
- 7 Conclusion

Why Assumption 1 is Stronger Than Desired

Assumption 1 requires: $F_e(d) \geq F_{\min} > 0$.

This means even the *highest price* must be accepted with positive probability by both groups.

In practice: a seller may not know which price is the true maximum acceptable price. Setting v_d too high would violate the assumption.

Why it's needed technically:

1. Ensures $\mathbf{1}^\top F_e \pi^e > 0$, so accepted-price ratio is well-defined.
2. Makes $S(\pi, F_1, F_2)$ Lipschitz — required for the w -grid approximation.
3. Allows accurate estimation of all $F_e(i)$.

Possible Relaxation

Open problem: Does there exist an algorithm that:

- ▶ Works for $F_e(i) > 0$ generally (not needing $F_e(d) > 0$)?
- ▶ Maintains optimal regret and unfairness rates?

The paper conjectures such an algorithm may exist but the proof techniques would need substantial modification.

Practical fix: Set v_d to a price you know some customers will accept, even if not the absolute maximum. Then reduce v_d until $F_e(d) \geq F_{\min}$ is plausible.

Anytime Fairness vs. Asymptotic Fairness

FPA's guarantee: $S(\pi_t; F_1, F_2) \rightarrow 0$ as $T \rightarrow \infty$ (asymptotically fair).

But: FPA **cannot** guarantee $S(\pi_t) = 0$ at every round t , since F_1, F_2 are unknown.

Practical benefit of epochs: Each policy runs for a full batch of $\tau_k/|A_k| = \Omega(\sqrt{T} \cdot 2^k)$ rounds. This is long enough for customers to observe and compare their average prices. So fairness is *felt* by customers within each epoch.

If the policy changed every round, customers would see no stable pattern and cannot assess fairness.

Relaxed Fairness Formulation

Instead of $S(\pi^*) = 0$, consider:

$$\pi_{\delta,*} = \arg \max_{\pi \in \Pi} R(\pi; F_1, F_2) \text{ s.t. } U(\pi) = 0, S(\pi) \leq \delta$$

Key relation (Lemma 14):

$$R(\pi^*) \leq R(\pi_{\delta,*}) \leq R(\pi^*) + L\delta$$

Known results for extreme cases:

- ▶ $\delta = 0$: both regret and unfairness rates are $\Theta(\sqrt{T})$.
- ▶ $\delta \geq 1$: reduces to unconstrained pricing, optimal regret $\Theta(\sqrt{T})$, unfairness = 0.
- ▶ $\delta = O(1/\sqrt{T})$: still achieves $O(\sqrt{T})$.

Conjecture: For $\delta > 1/\sqrt{T}$, optimal unfairness $\approx \Theta(1/(\sqrt{T}\delta))$.

Natural Extension to $G > 2$ Groups

Multi-group fairness: Define unfairness as the sum of pairwise unfairness over all $\binom{G}{2}$ pairs:

$$U_G(\pi) = \sum_{e < e'} |v^\top \pi^e - v^\top \pi^{e'}|$$

$$S_G(\pi) = \sum_{e < e'} S(\pi^e, \pi^{e'}; F_e, F_{e'})$$

Extension of FPA:

- ▶ Lengthen each epoch by a factor of $G/2$.
- ▶ Keep all other steps identical.
- ▶ Upper regret bound: $\tilde{O}(G^3 \sqrt{T} d^{2/3})$.
- ▶ Still optimal w.r.t. T up to $\log \log T$ factors.

Continuous Price Space

The paper restricts to finite V . What if prices can be any $v \in [0, 1]$?

Challenge:

- ▶ Optimal fair policy may be a continuous distribution.
- ▶ Even with known F_e , solving the optimization is harder.
- ▶ Online learning further amplifies difficulty.

Potential approach: Continuous distribution discretization (Xu and Wang, 2022), but this leads to *exponential time complexity*.

Open problem: Is there a polynomial-time algorithm for doubly-fair pricing on the continuous space with $\tilde{O}(\sqrt{T})$ guarantees?

Can We Trade Off One Fairness for the Other?

Conjecture of the paper: It is *not* productive to trade substantive fairness for procedural fairness.

Reasoning: The lower bound on substantive unfairness (Theorem 9) arises from **indistinguishability between environments**. Allowing procedural unfairness would not make the two environments more distinguishable — since the statistical difficulty is about inferring the optimal *acceptance* structure, not just the offered prices.

In other words: Setting unfair prices intentionally would not substantially speed up the learning process. So the fundamental $\Omega(\sqrt{T})$ substantive unfairness remains.

Practical implication: A seller does not gain much by relaxing procedural fairness (offering different average prices). The learning bottleneck is in discovering the *accepted-price structure*, which requires time regardless of whether offered prices are fair.

- 1 Background & Motivation
- 2 Motivating Example
- 3 Problem Setup
- 4 The FPA Algorithm
- 5 Regret and Unfairness Analysis
- 6 Discussion and Extensions
- 7 Conclusion**

What This Paper Achieves

1. **New fairness framework:** Two distinct, well-motivated fairness notions for pricing (procedural and substantive) — the second being new to the literature.
2. **Necessity of randomization:** Proved that deterministic doubly-fair pricing can be strictly revenue-inferior; random policies are mathematically essential.
3. **FPA algorithm:** First dynamic pricing algorithm that simultaneously satisfies both fairness constraints while learning from binary feedback, achieving $\tilde{O}(\sqrt{T})$ regret.
4. **Near-optimality:** Matching lower bounds show FPA is information-theoretically optimal in both regret and substantive unfairness up to $\log \log T$ factors.
5. **Computational efficiency:** Reduction to a sequence of LPs via the w -trick; polynomial time per epoch.

Open Problems I: Assumptions and Dimension

1. Weaker Demand Assumptions

The analysis assumes a lower bound on acceptance probabilities, so every price retains enough statistical signal. A natural next step is to handle prices that may be rarely accepted.

- ▶ Can FPA remain stable when some $F_e(v)$ is very small?
- ▶ What exploration rule prevents division-by-small-probability effects in substantive fairness?
- ▶ Can guarantees depend on an adaptive confidence radius instead of a global minimum acceptance rate?

2. Sharper Dependence on d

Current upper bounds scale as $\tilde{O}(\sqrt{T} d^{3/2})$, while the regret lower bound is only $\Omega(\sqrt{dT})$.

- ▶ Is the extra factor in d an artifact of the proof?
- ▶ Can confidence sets or LP structure close the $d^{3/2}$ vs. $d^{1/2}$ gap?
- ▶ Is there a dimension-adaptive version of FPA whose cost depends only on prices that are statistically relevant?

Main technical challenge: improve learning guarantees without losing either fairness constraint, especially under sparse demand or large price grids.

3. Continuous Price Spaces

The paper works with a finite price set V . Real platforms often choose prices from a continuum, e.g. $v \in [0, 1]$.

- ▶ What is the right analogue of randomized fair policies over continuous prices?
- ▶ Can the w -trick be made computationally efficient without exponential discretization?
- ▶ Is $\tilde{O}(\sqrt{T})$ still achievable in polynomial time?

4. More Than Two Groups

Pairwise summation is mathematically convenient, but may not be the cleanest fairness notion for many groups.

- ▶ Can one define a global fairness metric that avoids $\binom{G}{2}$ pairwise constraints?
- ▶ How should fairness scale with small or overlapping demographic groups?
- ▶ Can algorithms avoid the current polynomial blow-up in G ?

Practical direction: extend doubly-fair pricing from a clean two-group, finite-grid model toward real marketplaces with many protected groups and flexible price menus.

Summary: Everything in One Slide

Component	Definition	Key Formula
Policy	Distribution over prices per group	$\pi = (\pi^1, \pi^2) \in \Delta^d \times \Delta^d$
Expected Revenue	Weighted avg of group revenues	$R = q\mathbf{v}^\top F_1 \pi^1 + (1 - q)\mathbf{v}^\top F_2 \pi^2$
Proced. Unfairness	Gap in offered-price avg	$U(\pi) = \mathbf{v}^\top \pi^1 - \mathbf{v}^\top \pi^2 $
Subst. Unfairness	Gap in accepted-price avg	$S = \left \frac{\mathbf{v}^\top F_1 \pi^1}{\mathbf{1}^\top F_1 \pi^1} - \frac{\mathbf{v}^\top F_2 \pi^2}{\mathbf{1}^\top F_2 \pi^2} \right $
Optimal Policy	Max revenue s.t. $U = S = 0$	$\pi^* = \arg \max_{\pi \in \Pi} R \text{ s.t. } S = 0$
Regret	Revenue loss vs. π^*	$\text{Reg}_T = \sum_t [R(\pi^*) - R(\pi_t)]$
FPA UB (Thm 6)	$\tilde{O}(\sqrt{T})$ regret & unfairness	$\text{Reg}_T, S_T = \tilde{O}(\sqrt{T} d^{3/2})$
Regret LB (Thm 8)	No algorithm can beat $\sqrt{dT}$	$\text{Reg}_T \geq \Omega(\sqrt{dT})$
Unfairness LB (Thm 9)	Optimal $\Rightarrow$ some unfairness	$S_T \geq \Omega(\sqrt{T})$ for opt. algs
w-Trick	Linearizes non-convex constraint	Fix accepted-price level w ; solve LP

Thank You

Questions & Discussion

Reference

Jianyu Xu, Dan Qiao, Yu-Xiang Wang.

Doubly Fair Dynamic Pricing.

Proceedings of AISTATS 2023, Valencia, Spain. PMLR Vol. 206.